

1. Administrative & Study Identification

Full Study Title: A Phase I, Single-Dose, Open-Label, Sequential, Randomised, Crossover Study to Assess the Relative Bioavailability of Different Subcutaneous Formulations of AZD6234 in Participants Living With Overweight or Obesity **Short Title/ Acronym:** AZD6234 Formulations Bioavailability Study **Protocol Number:** Not specified **NCT Number:** NCT07220954 **Sponsor Name:** AstraZeneca **Investigational Product:** AZD6234 (Amylin receptor agonist formulations) **Phase of Development:** Phase I **Medical Monitor:** AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective **Primary Objective:** To compare the blood levels (relative bioavailability and pharmacokinetics) of different subcutaneous formulations of AZD6234 administered as a single dose. **Secondary Objectives:** To evaluate the safety, tolerability, and side effects of the different formulations, and to assess the effect of the test medicine on the immune system response (immunogenicity).

2.2 Background & Rationale To evaluate and identify the optimal subcutaneous formulation of AZD6234, an amylin receptor agonist, for ongoing clinical development in weight management for individuals living with overweight or obesity. Understanding the bioavailability and safety profiles of Formulations 1, 2 (low and high concentrations), and 3 will guide subsequent Phase II/III trials.

3. Study Design & Methodology

3.1 Design Framework **Study Type:** Interventional **Control Type:** Active Comparator (Crossover) **Blinding:** Open-label **Randomization:** Randomised, Sequential

3.2 Planned Enrollment & Duration **Target** \$N\$: 21 healthy participants **Number of Sites:** 1 site (Nottingham, United Kingdom) **Estimated Study Start:** 10-Nov-2025 **Estimated Primary Completion:** 24-Mar-2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Healthy males or non-pregnant, non-lactating females aged 18 to 55 years inclusive. 2. Body Mass

Index (BMI) of 25.0 to 35.0 kg/m² inclusive, and body weight $\geq$ 50 kg.

4.2 Exclusion Criteria 1. Pregnant or breastfeeding females. 2. Any significant clinical condition, prior treatments, or immune system disorders that would interfere with study participation, safety tests, or pharmacokinetic assessments as determined by the investigator.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Single dose of a specific AZD6234 formulation (Formulation 1, Formulation 2 low concentration, Formulation 2 high concentration, or Formulation 3) administered without food across 4 separate treatment periods. **Route of Administration:** Subcutaneous (SC) injection into the abdomen (stomach). **Duration of Treatment:** Total study duration of up to 19 weeks.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care consistent with Phase I confinement protocols. **Prohibited:** Food prior to dosing (administered in a fasted state); concurrent investigational medications or therapies known to impact weight or gastric emptying.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
1	Screening	Prior to Period 1 Informed Consent, Medical History, BMI verification, Baseline Labs
2	In-Clinic Stays	4 Treatment Periods 6 nights in the clinic per occasion; Single dose administration (fasted), blood/urine PK sampling, safety monitoring
3	Outpatient Visits	Between Periods Up to 10 outpatient visits for continued PK/safety monitoring and washout clearance
4	End of Study	Up to Week 19 Final safety tests, immune response assessment, exit interview

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Relative bioavailability determined by the measurement of the amount of AZD6234 and its breakdown products in the blood (Pharmacokinetics: Area Under the Curve [AUC] and Maximum Concentration [C_{max}]). **Measurement Method:** Central Laboratory Analysis of intensive blood and urine samples.

7.2 Secondary & Exploratory Endpoints Incidence and severity of side effects and Adverse Events (Safety and Tolerability). Effect

of the AZD6234 formulations on the immune system response (Immunogenicity).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Intensive monitoring by clinical staff during the 6-night inpatient stays and follow-up during the 10 outpatient visits. **Serious Adverse Events (SAEs):** Standard reporting timeline (typically 24 hours) to the sponsor. **Data Monitoring Committee (DMC):** Internal safety review by the Principal Investigator and AstraZeneca medical monitors.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Pharmacokinetic (PK) Analysis Set for primary relative bioavailability outcomes; Safety Set for all participants receiving at least one dose. **Sample Size Justification:** Target $N=21$ participants, appropriately powered to evaluate intra-subject variability in a crossover design for bioavailability endpoints. **Handling of Missing Data:** Standard Phase I PK methodologies (e.g., omitting subjects with missing critical PK timepoints from the primary PK analysis set).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Continuous onsite monitoring during inpatient confinement periods. **Audit Trail:** Standard Phase 1 eCRF locking, source data verification against clinic records, and pharmacokinetic data clarification forms.

11. Study Identifiers & Governance

Primary ID: NCT07220954 **Registry Number:** NCT07220954 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** A Study to Assess the Relative Bioavailability of Different Subcutaneous Formulations of AZD6234 **Official Regulatory Title:** A Phase I, Single-Dose, Open-Label, Sequential, Randomised, Crossover Study to Assess the Relative Bioavailability of Different Subcutaneous Formulations of AZD6234 in Participants Living With Overweight or Obesity

12. Clinical Framework (Overweight / Obesity Specific)

Primary Condition: Overweight and Obesity **Diagnosis Threshold:** BMI of 25.0 to 35.0 kg/m² inclusive **Medical Methodology:** Amylin Receptor Agonist Therapy (AZD6234) **Optimization Target:**

Pharmacokinetic and bioavailability assessment to select optimal formulation

13. Study Procedures & Methodology

Management Pathway: Phase I intensive pharmacokinetic clinical trial pathway. **Education Protocol (HCP):** Protocol-specific training for Phase I inpatient unit staff. **Education Protocol (Patient):** Clinic confinement instructions, intensive sampling schedule consent, and fasting requirements. **Quality Audit Mechanism:** Strict control of meal times, dosing administration, and precise timing of blood/urine collections for PK analysis.

14. Observation & Follow-Up Schedule

Total Duration: Up to 19 weeks **Onsite Visit Cadence:** 4 distinct inpatient periods (6 nights each) and up to 10 outpatient visits **Remote Monitoring Frequency:** N/A (intensive in-clinic and outpatient tracking) **Checklist Review Interval:** Continuous AE and tolerability monitoring during every clinical interaction

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				